

ENBC332: Homework #7

Due: Monday, 11/10/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 40

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: “problem1_fullname.R”. Start with the problem number and followed by your full name, e.g., “**problem1_jimmyazarnoosh**”.

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:
e.g., calculating range.

```
# -----  
Data <- trees$Height  
my_range_func <- function(Data){  
  Range <- max(Data) - min(Data)  
  return(Range)  
}  
# -----
```

Dependent and Independent Samples

Problem 1. This problem helps you understand hypothesis testing using a paired sample design, focusing on calculating and interpreting p-values with R. (20 points)

You are studying the effects of a specialized diet program on weight loss. You collected data from 20 participants, measuring their weight (in kg) before and after following the diet for 12 weeks. The data is as follows:

- **Weight Before (in kg):** 78, 85, 92, 68, 74, 88, 81, 76, 95, 70, 79, 82, 91, 72, 80, 83, 77, 89, 84, 73
- **Weight After (in kg):** 75, 81, 89, 66, 71, 84, 78, 73, 90, 68, 76, 79, 87, 70, 77, 80, 74, 86, 80, 71

Write R Code: (10 points)

- Use R to perform a paired t-test to analyze the data and determine if the diet program resulted in significant weight loss.
- Calculate the p-value and interpret it in the context of this problem. Use a significance level of 0.05.

Conceptual questions:

1. State the null and alternative hypotheses to determine if there is a significant difference in the participants' weight before and after the diet. (4 points)
2. Explain what the p-value indicates about the effectiveness of the diet program. (3 points)
3. State whether you reject or fail to reject the null hypothesis and what that means in practical terms. (3 points)

Problem 2. This problem helps you understand hypothesis testing using two-sample t-test, specifically interpreting and calculating p-values, using R for data analysis. (20 points)

You want to analyze the effectiveness of a new drug intended to lower blood pressure. You collected data from a clinical trial where 50 patients were randomly assigned to either a treatment group (receiving the drug) or a control group (receiving a placebo). After 8 weeks of treatment, you measured the change in systolic blood pressure (in mmHg) for each participant.

The data is given below:

- **Treatment Group:** -12, -8, -10, -5, -7, -13, -9, -11, -6, -10, -15, -8, -11, -14, -9, -13, -8, -12, -10, -7, -11, -9, -14, -10, -12
- **Control Group:** -3, -4, -2, -5, -6, -3, -4, -5, -2, -1, -3, -4, -6, -3, -2, -4, -5, -2, -3, -1, -4, -3, -6, -2, -4

Write R Code: (10 points)

- Perform an independent two-sample t-test using R to determine if there is a significant difference between the two groups.
- Calculate the p-value and interpret its meaning in the context of this problem. Use a significance level of 0.05.

Conceptual questions:

1. State the null and alternative hypotheses for comparing the mean change in systolic blood pressure between the treatment and control groups. (4 points)
2. Explain what the p-value indicates about the effectiveness of the drug. (3 points)
3. Discuss whether you would reject or fail to reject the null hypothesis and what that decision implies about the drug's efficacy. (3 points)